





TestAI

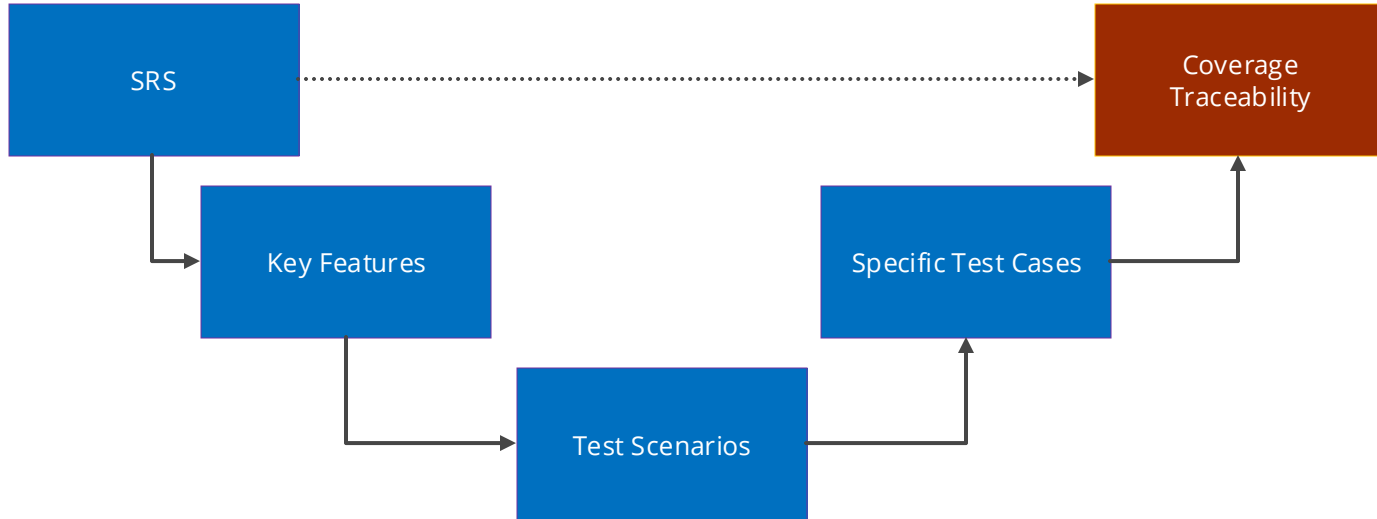
AI Based Verification



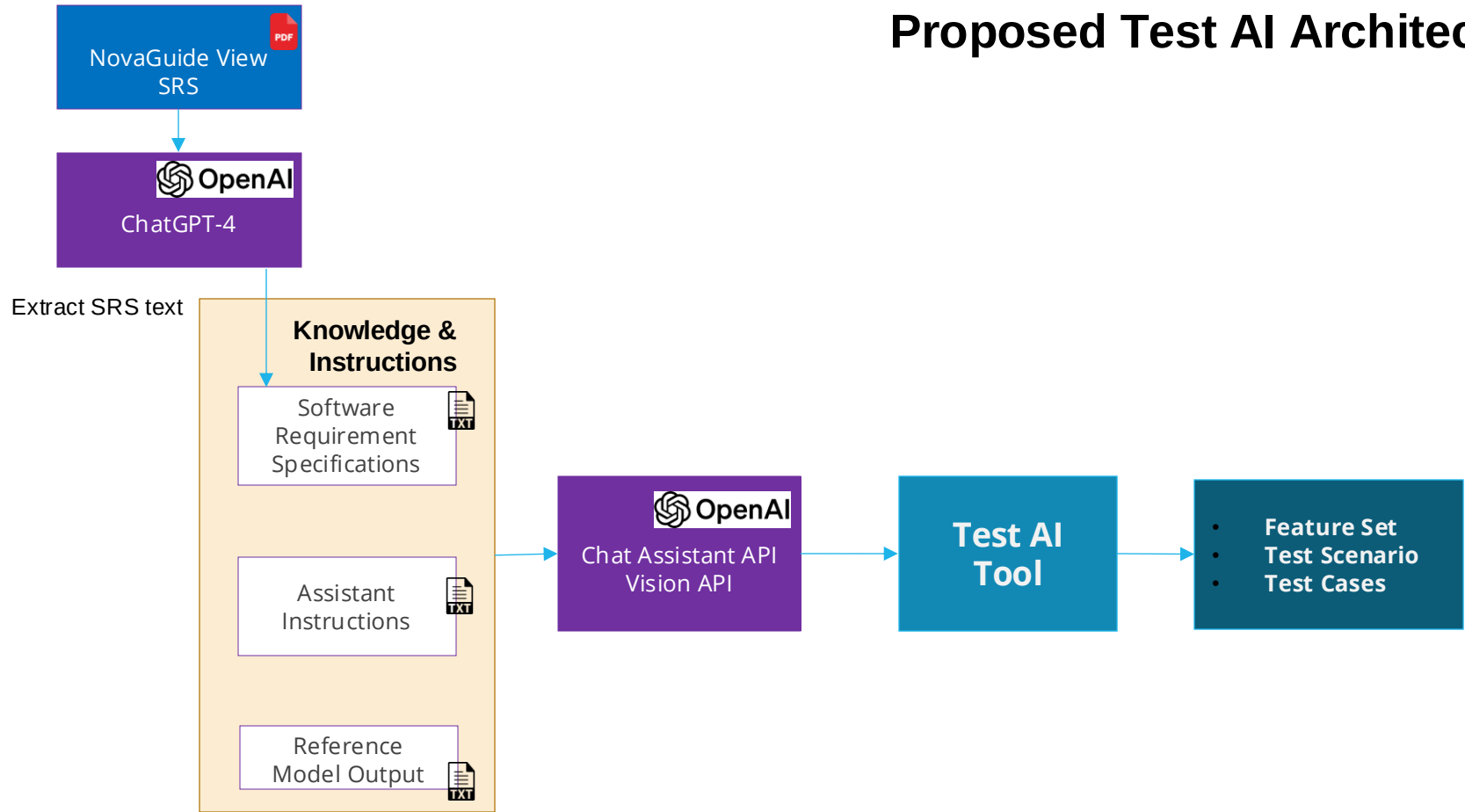
Introduction

- **Objective:** Leverage AI to effectively test the software product
- **Status:** Proof of concept completed
- **Issues:** Hallucinates features and definitions
- **Recommendations:**
 - It's a powerful and efficient test case generation tool
 - Explore traceability and test coverage

Workflow



Proposed Test AI Architecture



Product Features

TestAI

Name

TestAI

asst_XkYFNyuaEufxjH6kok2HzrF5

Instructions

You are a Software testing specialist, the SRS for the product is attached in the files, based on the SRS, if test scenarios or test cases are to be created, you will be prompted, please take the reference of SRS to create the test scenarios and test cases

Model

gpt-4-1106-preview

TOOLS

Functions

Code interpreter

Retrieval

FILES

DC-01937 rev H_NGV SRS, PL23.1-NGV4.0.0.docx

THREAD thread_WFnK49oqgpFjRt9RycpxgQ01

77460 tokens

User

List and detail the primary functions the software performs. Include any unique or innovative features that set the product apart. Explain how the core functions contribute to solving user problems

Run instructions

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TestAI

The software, NovaGuide View PWA version 3.0.3, includes a range of features designed to support medical practitioners in monitoring and reporting on Transcranial Doppler (TCD) exams. The primary functions are:

- Live Streaming:** Mirror mode monitoring for NG Exam's TCD Exam parameters [1] .
- Reporting Enhancements:** Provides 2D and 3D Circle of Willis (CoW) with interactive visualizations [2] .
- Medical Interpretation Entry:** Allows medical practitioners to enter their interpretations and notes directly into patient reports, which can then be saved as DICOM and sent back to the hospital system [3] .
- Threshold Advisories:** Advisories based on changes in TCD exam's parameters are automatically generated [4] .
- Role-based Access:** Features role-based templates through an IT Manager's portal, authorizing specific roles to access app features [5] .
- Cloud-based Computations:** Quantum Cascade Harmonic (qCh) based computations in the cloud for threshold advisories [6] .
- Remote Access and Storage:** Capability to upload, store, and view TCD exam data remotely as well as to search across multiple patients and exams [7] .
- Reporting and Record Storage:** Real-time monitoring with advisories on a remote app or device, including the ability to

Test Scenarios



TestAI

Name

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Model

gpt-4-1106-preview

TOOLS

Functions 

Code interpreter 

Retrieval 

FILES 

DC-01937 rev H_NGV SRS, PL23.1-NGV4.0.0.docx

THREAD thread_WFnK49oqgpFjRt9RycpxgQ01

77460 tokens

User

Derive test scenarios to evaluate the security measures outlined in the SRS. Include scenarios for authentication, authorization, data encryption, and protection against common security threats. Test for vulnerabilities and ensure compliance with security standards.

TestAI

To derive test scenarios to evaluate the security measures of NovaGuide View PWA as described in the SRS (Software Requirements Specification), we would create test scenarios around specific security requirements, including authentication, authorization, data encryption, and protection against common security threats. Below are sample test scenarios for each category:

1. Authentication:

- **Test Scenario 1: Valid User Login**
 - Attempt to log in with valid user credentials.
 - Expected result: Authentication is successful; the user accesses the application.
- **Test Scenario 2: Invalid User Login**
 - Attempt to log in with invalid user credentials.
 - Expected result: Authentication fails; the user is denied access to the application.
- **Test Scenario 3: Account Lockout Mechanism**
 - Attempt to log in with incorrect passwords multiple times.
 - Expected result: Account becomes locked after a configured number of failed attempts to prevent brute force attacks.

2. Authorization:

- **Test Scenario 4: Role-Based Access Control**

Test Cases

TestAI

Name

TestAI

asst_XkYFNyuaEuFxfjH6kok2HzyF5

Instructions

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Model

gpt-4-t106-preview

TOOLS

Functions

Add

Code interpreter

Off

Retrieval

On

FILES

Add

DC-01937 rev H_NGV SRS, PL23.1-NGV4.0.0.docx

THREAD thread_WFnK49oqgpFjRt9RycpxgQ01

89378 tokens

User

Develop test cases to validate the streaming functionality of the web app. Specify steps for testing individual features of streaming listed in the SRS

Run instructions

You are a Software testing specialist, the SRS for the product is attached in the files, based on the SRS, if test scenarios or test ...

TestAI

To validate the streaming functionality of the NovaGuide View PWA as outlined in the SRS, we can develop several test cases covering various aspects of the live streaming features. Below are some examples of test cases based on the extracted SRS information:

Test Case 1: Live Streaming Access and Quality

- Objective: Validate that authorized users can access live streaming and observe high-quality exam parameters including Depth changes, Envelope, Gain, Sample size, and Scale.
- Steps:
 - 1 Log in with an authorized NovaGuide View account.
 - 2 Initiate a live stream from a NovaGuide exam.
 - 3 Verify that exam parameters are observable and check for Depth changes, Envelope, Gain, Sample size, and Scale during the live stream.
 - 4 Confirm that the sound quality in the live stream is audibly accurate, especially the high-frequency sounds, and compare it with the TCD audio data from the NG2 device.
- Expected Result: The live stream is accessible, exam parameters are visible and modifiable in real-time, and the sound quality is comparable to the NG2 device.

Session based test case generation

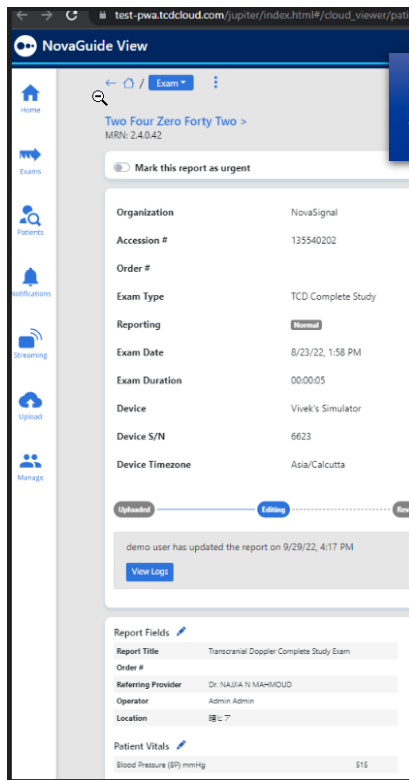
The screenshot shows a chatbot conversation. The assistant asks for test-related questions on Jupiter. The user requests a list of scenarios for testing Jupiter. The assistant responds with a list of scenarios, which are highlighted in a green box. The scenarios are:

- 1. NG View Authorization (PL23.1_NGV_1.1)**
 - Scenario: Verify user authentication using hospital's authentication system (Microsoft Active Directory, SAML, or Google).
 - Scenario: Validate admin user access and features accessible like User Profile Setup, Deletion, and User Access Privileges.
- 2. Live Streaming Enhancement on NG View (PL23.1_NGV_1.2)**
 - Scenario: Check the ability to view a live exam with consistent sound output frequency response and dynamic range.
 - Scenario: Validate that TCD audio data spectrum is the same between NG2 device and NGV.
- 3. Notification Creation while in Live Streaming on the PWA App (PL23.1_NGV_1.3)**
 - Scenario: Test the creation of new conditions for notifications during Live Streaming mode.
 - Scenario: Verify the ability to choose notification means (SMS, email).
- 4. Medical Record and Notes Security in NG View (PL23.1_NGV_1.4)**
 - Scenario: Ensure medical records and notes entered into NG View are securely stored and accessed.

The screenshot shows a chatbot conversation. The user asks for test cases for a specific test scenario. The assistant responds with detailed test cases for the scenario "Live Streaming Enhancement on NG View (PL23.1_NGV_1.2)". The test cases are highlighted in a green box. The test cases are:

- Test Case 1: Live Streaming connectivity and quality**
 - Objective:** To verify that the Live Streaming feature connects successfully and streams with the required quality.
 - Preconditions:** User is authenticated and has the necessary permissions to access Live Streaming.
 - Steps:**
 - Log into the NG View with valid credentials.
 - Navigate to the Live Streaming section.
 - Select an active exam to view live.
 - Expected Result:**
 - The system should authenticate the user successfully.
 - The live streaming should start without significant delay.
 - The video and audio quality should meet predefined quality metrics (e.g., clear visuals, no lag, consistent sound output).
 - Actual Result:** (To be filled after execution)
 - Pass/Fail:** (To be marked after execution)
- Test Case 2: Live Streaming audio data consistency**
 - Objective:** To ensure TCD (Transcranial Doppler) audio data continuity and consistency during Live Streaming between the NG2 device and NGV.
 - Preconditions:** Live Streaming is in progress.

General



From screenshots identify the features to be tested

```
print(response.json()[\"choices\"][0][\"message\"][\"c
```

```
with open(image_path, 'rb') as image_file:
    return base64.b64encode(image_file.read()).decode('utf-8')

# Path to your image
image_path = "2022-09-29 16_20_43-Window.png"

# Getting the base64 string
base64_image = encode_image(image_path)

headers = {
    "Content-Type": "application/json",
    "Authorization": f"Bearer {api_key}"
}

payload = {
    "model": "gpt-4-vision-preview",
    "messages": [
        {
            "role": "user",
            "content": [
                {
                    "type": "text",
                    "text": "What are the important aspects of this screen that would have to be tested?"
                }
            ]
        }
    ]
}
```

When testing a screen like this in a medical or clinical software application, there are several important aspects you would want to consider:

- Functionality:**
 - Verify that all interactive elements on the screen function as intended (e.g., buttons, checkboxes, links, and dropdown menus).
 - Ensure that the "Mark this report as urgent" checkbox toggles correctly and triggers the appropriate workflows.
 - Confirm that the editing status progresses correctly from "Uploaded" to "Editing" and any other states as appropriate.
 - Test the "View Logs" button to ensure it shows the correct log information.
- Data Validation & Accuracy:**
 - Check that all displayed information matches the data entered or imported, ensuring accuracy for patient identifiers, dates, and exam details.
 - Validate input fields to confirm they handle data formats correctly and prevent entry of invalid data.
- User Interface and Layout:**
 - Ensure that the layout is consistent with the application's design standards.
 - Confirm that text is readable and that controls are accessible and clearly labeled.
 - Test responsiveness of the design, if applicable, to ensure the layout adjusts correctly for different screen sizes and orientations.
- Navigation:**
 - Test that navigation elements like the back arrow and breadcrumbs work correctly and lead to the expected parts of the application.
 - Verify the functionality of the side menu and check that switching between sections works properly.
- Permissions and Access Control:**

Specific

The screenshot shows the NovaGuide View application interface. The top navigation bar includes a home icon, a search icon, and a dropdown menu with 'Exam' selected. The main content area displays details for 'Two Four Zero Forty Two' (MRN: 24042). A sidebar on the left contains icons for Home, Exams, Patients, Notifications, Screening, Upload, and Manage. The main content area includes a 'Mark this report as urgent' checkbox, a 'Report' status, and a 'View Logs' button. Below this, there is a 'Report Fields' section with fields for Report Title, Order #, Referring Provider, Operator, Location, and Patient Vitals.

Generate specific test cases

```
# Path to your image
image_path = "2022-09-29 16_20_43-Window.png"

# Getting the base64 string
base64_image = encode_image(image_path)

headers = {
    "Content-Type": "application/json",
    "Authorization": f"Bearer {api_key}"
}

payload = {
    "model": "gpt-4-vision-preview",
    "messages": [
        {
            "role": "user",
            "content": [
                {
                    "type": "text",
                    "text": "Can you write a test case in detailed how to test the most important aspect in this screen?"
                },
                {
                    "type": "image_url",
                    "image_url": {
                        "url": f"data:image/jpeg;base64,{base64_image}"
                    }
                }
            ]
        }
    ]
}
```

```
print(response.json()[0]["choices"][0]["message"]["content"])
✓ 0.0s

Test Case: Verify the Functionality of the "Mark this report as urgent" Checkbox

Test Case ID: TC_NV_001
Prepared By: [Tester Name]
Date Prepared: [Current Date]
Preconditions: User must be logged into the NovaGuide application and have access to the report page.

Test Steps:
1. Navigate to the "Exams" section and select a particular exam to view its details.
2. Locate the "Mark this report as urgent" checkbox on the exam report page.
3. Observe the default state of the checkbox upon loading the report page. Expected: The checkbox should not be marked by default.
4. Click on the checkbox to mark the report as urgent.
5. Verify that upon clicking, a visual indicator (such as a checkmark or a change in color) appears within the checkbox.
6. Confirm that any related visual elements or messages that should appear or change upon marking a report as urgent are displayed correctly (e.g., "Urgent" badge, notification to the medical staff, etc.).
7. Uncheck the "Mark this report as urgent" checkbox to see if it correctly represents the non-urgent state.
8. Save or submit the report (if such a step is required) and navigate away from the page.
9. Return to the same report to verify that the state of the "Mark this report as urgent" checkbox is retained as you left it (either marked as urgent or not urgent).
10
```

SRS Verification

1	Complete	All items needed for specify the solution to the problem have been included
2	Correct	Each item is free from error
3	Precise, unambiguous, and clear	Each item is exact and not vague; there is a single interpretation; the meaning of each item is understood; the specification is easy to read
4	Consistent	No item conflicts with another item in the specification
5	Relevant	Each item is pertinent to the problem and its solution
6	Testable	During program development and acceptance testing, it will be possible to determine whether the item has been satisfied
7	Traceable	Each item can be traced to its origin in the problem environment
8	Feasible	Each item can be implemented with the available techniques, tools, resources, and personnel, and within the specified cost and schedule constraints
9	Free of unwarranted design detail	The requirements specifications are a statement of the requirements that must be satisfied by the problem solution, and they are not obscured by proposed solutions to the problem
10	Manageable	The requirements are expressed in such a way that each item can be changed without excessive impact on other items

Summary

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NeuraSignalTM